Characteristics of the Skeletal Biconical Antenna as Used for EMC Applications

Simon M. Mann and Andrew C. Marvin, *Member, IEEE*

***Abstract*—The Numerical Electromagnetics Code (NEC) is used to simulate the radiators of a skeletal biconical antenna and the model is optimized to return a drive point impedance as close as possible to measured values. The horizontally polarized model is then illuminated in free space and above a ground plane to deduce its plane-wave antenna factors and their variations as functions of height. It is shown that changes in effective length have to be considered in addition to changes in impedance mismatch when calculating antenna factor variations with the biconical antenna, although only the latter are necessary with a resonant dipole. The measured scattering parameters of the balun of the biconical antenna are then incorporated with the NEC simulation to give a complete model of the real antenna.**

I. Introduction

CONVENTIONAL approaches to the calibration of antennas for EMC applications involve measurements made over very expensive high-quality open-area test sites, but because they also rely upon propagation models and concepts such as normalized site attenuation [1], [2] which themselves have limited validities [3], typical quoted uncertainty budgets are not better than ± 2 dB.

Computer simulation is capable of yielding highly accurate antenna factor predictions, but it is difficult to have confidence in these unless they can be validated against measurement. The problem with the use of computer codes is therefore one of validation. In this paper, the NEC computer code [4] is used and antenna factor predictions are indirectly validated by demonstrating close agreement between the measured and predicted drive-point impedances of an antenna.

Antenna factors are known to vary as functions of antenna height above a ground plane due to mutual coupling. The extent of these variations has been calculated by various authors for resonant dipoles [1], [3] by a consideration of the change in mismatch between the antennas and their connecting transmission line. In this paper variations in the antenna factor of the resonant dipole and biconical antenna are calculated by a rather different approach which is able to consider changes in the effective length of an antenna in addition to changes in mismatch.

II. Calculation of Antenna Factors

Calibration of an antenna for EMC applications involves the determination of its antenna factors, which relate the received voltage to the illuminating field, i.e.,

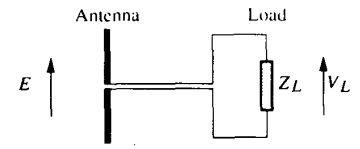


Fig. 1. A general receiving antenna.

$$AF = \left|\frac{E}{V_L}\right| \tag{1}$$

at a given frequency. Antenna factors are usually expressed in decibels so that they may be simply added to the received voltage in dBμV to yield the electric field strength in dBμV/m.

For a general receiving antenna connected to a load impedance, Z_L, as shown in Fig. 1, it is possible to show that the antenna factor is given by

$$AF = \left|\frac{E}{V_T}\left(1 + \frac{Z_T}{Z_L}\right)\right| \tag{2}$$

where V_T and Z_T are the EMF induced in the antenna and its drive-point impedance. The reciprocal of the first term in (2) is often known as the effective length of the antenna and the term in brackets describes the impedance mismatch between the antenna and its load.

A typical example of the use of (2) is for calculation of the antenna factor of a resonant dipole. This antenna is slightly less than half a wavelength long and its current is known to be distributed approximately sinusoidally over its length. If the antenna is assumed to be exactly half a wavelength long and to have a perfectly sinusoidal current distribution, its drive-point impedance can be shown to be equal to 73 Ω and its effective length can be calculated as $l_e = \lambda/\pi$, where λ is the wavelength. Under these conditions, (2) gives the antenna factor of the resonant dipole driving into a 50 Ω load as

$$AF = (20 \log_{10}(f_M) - 31.77)\ \text{dB} \tag{3}$$

where f_M is the frequency in megahertz. This formula has been shown to give results within 0.02 dB of NEC for real antennas such that $2\ln(2l/a) > 8$, where a is the radius and l is the half-length of an antenna [5].

Antennas other than the resonant dipole have more complicated current distributions and therefore there is no simple approach to the determination of their effective lengths and drive-point impedances. For antennas such as the skeletal biconical antenna and the logarithmic periodic antenna, it is

Manuscript received September 27, 1993; revised March 19, 1994.
S. M. Mann is with the National Radiological Protection Bourd, Chilton, Didcot, OXON, OX11 0RQ, United Kingdom.
A. C. Marvin is with the Department of Electronics, University of York, York, YO1 5DD, United Kingdom.
IEEE Log Number 9404479.

0018-9375/94$04.00 © 1994 IEEE

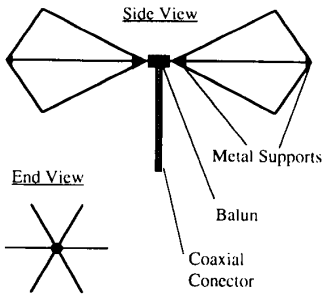


Fig. 2. The biconical antenna.

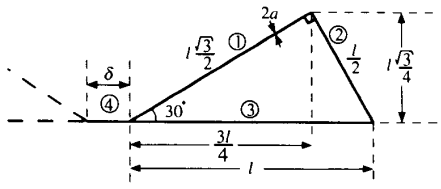


Fig. 3. The biconical antenna.

possible to use the NEC computer code to analyze model antennas and obtain predictions for the performances of the real antennas.

III. The Biconical Antenna

The biconical antenna is a broadband antenna that is usually specified for use over the VHF frequency range, i.e., 30–300 MHz, although certain slightly larger models are specified for the 20–200 MHz range. Most antenna manufacturers produce biconical antennas for EMC applications and these all have similar wire radiating structures, which are mounted upon a support containing a balun. Some of these support structures are plastic and others are metal; the Schwarzbeck BBA9106 antenna, which forms the subject of this paper, has a metallic support.

A. Physical Characteristics

The Schwarzbeck biconical antenna is similar to most other biconical antennas in that its structure consists of wire cages mounted either side of a support containing its balun. Fig. 2 shows this structure. Each cone of the antenna is formed from six elbow-shaped wires arranged around a single straight wire along the central axis. The angle between each bent cone wire and the central wire is 30° and the angle at each bend is 90°; therefore, the entire structure of each cone may be defined by a single cone-length dimension l, as shown in Fig. 3.

In addition to the cone length, the radius of the cone wires and the separation of the cones at the antenna feed are required to define the complete radiating structures of the antenna. Their measured values were as follows:

Cone separation $\delta = 87$ mm
Cone length $l = 603.5$ mm
Wire radius $a = 3$ mm.

TABLE I
Initial Segmentation Scheme Used for the Skeletal Biconical Antenna

Wire Number	Length mm	No. of segments	Segment lengths $/\lambda$ at 300 MHz
1	522.646	10	0.0523
2	301.750	5	0.0604
3	603.500	11	0.0549
4	87.000	3	0.0290
Total number of segments = 205			

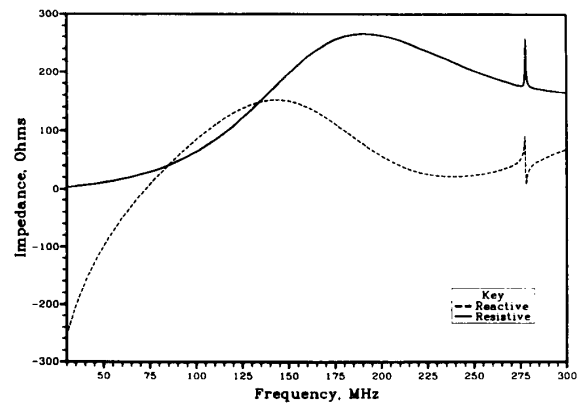


Fig. 4. Predicted self-impedance components for the skeletal biconical antenna.

B. Physical NEC Model

The measured dimensions of the biconical antenna were used to develop a NEC model with the initial segmentation scheme shown in Table I. A segment length corresponding to approximately $\lambda/20$ at 300 MHz was decided to be appropriate to ensure reasonable accuracy and a not too computationally demanding number of segments.

It was not possible to develop a physical model of the excitation region of the antenna without compromising NEC's own design rules for modeling structures with thin wires. Because of this, the region between the cones of the antenna was simply modeled as an 87 mmlong cone-linking wire, having the same radius as all of the other wires in the model, i.e., 3 mm. Three segments were used to model this wire so that the middle one could be used to specify a voltage excitation for the antenna.

C. Electrical Characteristics

The physical NEC model was excited by a voltage source and simulations were carried out to return the drive-point impedance at 5 MHz intervals from 30 to 300 MHz. More data points were obtained to reveal the detailed behavior of the impedance components where they had a strong dependence upon frequency and the resulting predictions are shown in Fig. 4.

Fig. 4 shows that the impedance of the biconical antenna becomes purely resistive and equal to 26.97 Ω at 72.65 MHz. This is the resonance of the biconical antenna but it is not necessarily the frequency at which minimum SWR occurs

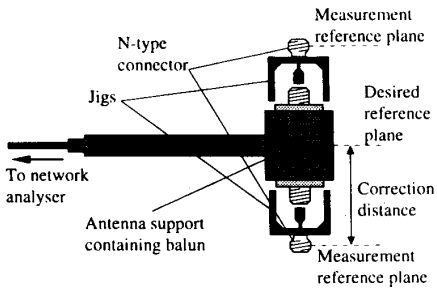


Fig. 5. Use of jigs to facilitate calibration of a network analyzer to measure the drive-point impedance of the biconical antenna.

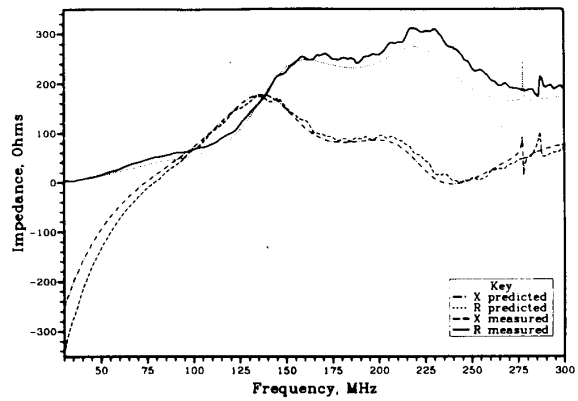


Fig. 6. Measured and predicted impedance components of the biconical antenna horizontally polarized at a height of 1.542 m above a ground plane.

when driven from a transmission line. At frequencies below resonance, the impedance becomes predominantly capacitive and increasing mismatch losses reduce the sensitivity of the antenna. Above resonance the antenna maintains an SWR of less than 5.6:1.

The impedance predictions show a sharply tuned resonant feature which causes the impedance components to be disturbed over a 3 MHz frequency range centered upon 277.77 MHz. The current on the central wire of each cone was found to increase markedly in the region of this cone resonance.

IV. Optimization of the NEC Model

The NEC model derived using the physical dimensions of the biconical antenna did not have the same physical structure as the real antenna, but represented an approximation formed from straight wires. There were three main differences between the model antenna and the real antenna:

1) The feed region of the model antenna was represented by a 6 mm diameter cone-linking wire, whereas the feed of the real antenna was contained inside a 40 mm metal cube which had coaxial penetrations through which each cone was fed.

2) The cones of the real antenna had bulky metal supports at either end which were not accounted for in the NEC model.

3) The outer cone wires of the real antenna had a finite bending radius at the widest point of the antenna, whereas those of the model had a sharp junction between two straight wires.

In order to deduce the effect of these approximations, the NEC model was optimized to return a drive-point impedance that agreed as closely with measurement as possible.

A. Measurement of Drive-Point Impedance

In order to measure the impedance of the radiators of the biconical antenna it was necessary to calibrate a network analyzer to establish a reference plane at their drive-point. This was accomplished by using jigs to mount N-type connectors on either side of the antenna support. By connecting pairs of open, short, and matched loads to the N-type connectors, the network analyzer was calibrated for an impedance measurement between the twin reference planes shown in Fig. 5.

The physical distance between each of the measurement reference planes and the desired reference plane was 75 mm, but 28 mm of this transmission line was filled with a dielectric having a velocity factor of 0.67. It was calculated that the equivalent free-space distance between the reference planes was 89 mm and therefore an appropriate phase correction was applied to the measured reflection coefficients to move the reference planes together and to the drive point.

The maximum frequency at which the impedance of the biconical antenna was measured was 300 MHz, corresponding to a wavelength of 1 m. With a wavelength as large as this, an error of 10 mm in the correction distance would give a phase error of less than 7.2° in the measured reflection coefficient.

It is important to note that, since the reference plane at the drive point was formed by connecting the two 50 Ω measurement reference planes in series, it had a characteristic impedance of 100 Ω and therefore the measured impedances were normalized to 100 Ω.

The antenna was mounted on a wooden tripod, horizontally polarized at a height of 1.542 m above the metal ground plane of an open-area test site. The resulting measured impedance components are shown, together with predictions for the physical model antenna above a perfect ground plane, in Fig. 6.

The measured and predicted impedance components in Fig. 6 show good qualitative agreement but their quantitative agreement is not very good. In particular, the predicted resistive components are in error by as much as 40 Ω around 225 MHz and the reactive predictions are a similar amount in error below 70 MHz.

B. Optimization of Model Dimensions

Fig. 6 shows that the predicted frequency of the cone resonance, i.e., 277.77 MHz, was too low and the actual frequency where it occurred was 287.3 MHz. It has been stated that the cone resonance is associated with currents on the central cone wire, it therefore follows that the cone length is the only dimension of the antenna which affects its frequency. It was found that reducing the cone length from its measured value of 603.5 mm to 585.275 mm moved the predicted cone resonance into agreement with measurement. It also improved the correlation between the predicted and the measured impedance components at all other frequencies.

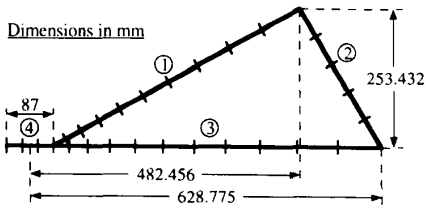


Fig. 7. Tapered segmentation scheme used for the cone wires.

TABLE II
SEGMENTATION SCHEMES USED FOR THE SKELETAL BICONICAL ANTENNA AND THEIR CORRESPONDING PREDICTED CONE RESONANT FREQUENCIES

Model No.	Number of segments				Cone Resonance
	Wire 1	Wire 2	Wire 3	Total	
1	4	2	5	87	281.76
2	6	3	8	129	285.52
3	8	4	10	169	286.30
4	10	5	12	209	286.70
5	15	7	18	317	287.23

The implication of having to perform the above correction to the cone lengths of the model antenna was that their electrical length was greater than that of the real antenna. This was because the bulky metal supports at either end of the real antenna reduced the length of the cavity inside its cones by around 18 mm. If a length correction of this kind were to be avoided with a NEC model of a biconical antenna, it would be necessary to incorporate the cone supports in the model.

Changing the length of the cone-linking wire was found to have only a very slight effect upon the impedance predictions, but eventually a separation of 95 mm was decided to give slightly better predictions than 87 mm.

C. Optimization of Segmentation

The number of segments used to model the cone-linking wire was found to have a critical effect upon impedance predictions. A variety of models with 3, 5, 7, and 9 segments for this wire were tried, and models with 5 segments were found to give the best predictions.

It is widely known that care must be taken when modeling the excitation region of any antenna with NEC, and particularly great care must be taken with the skeletal biconical antenna because there are two eight-wire junctions close to its excitation region. In order to allow the most accurate possible expansion of the current basis functions in this region, a tapered segmentation scheme was devised for the cone wires which gave their nearest segments to the excitation region the same lengths as those used for the cone-linking wire. This segmentation scheme is shown in Fig. 7 and was found to improve predictions.

The number of segments used to model the cone wires was reduced until the impedance predictions started to degrade. The first evidence of this was a gradual reduction in the frequency of the predicted cone resonance, as shown in Table II, which occurred because the tapered segmentation scheme caused undersegmentation of the cone wires towards the ends of the antenna.

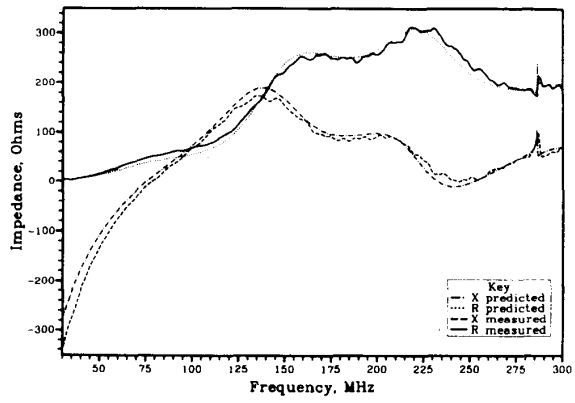


Fig. 8. Measured drive-point impedance components of the biconical antenna and predictions from the optimized NEC model.

The model used for the remainder of the work in this paper was formed with 169 segments and gave a predicted cone resonant frequency of 286.3 MHz. Slightly reducing the cone length would have returned the cone resonance to 287.3 MHz, but this was not done. Fig. 8 shows the impedance predictions from this model.

The impedance predictions from the optimal model show very close correlation with measurement at frequencies above 70 MHz. Below this frequency, the model antenna is slightly too capacitive and at 30 MHz the predicted capacitance is 17.93 pF whereas the measured capacitance is 15.39 pF. This difference is thought to arise because of interactions associated with the metal supports of the feed region of the real antenna which were not modeled.

V. ANTENNA FACTOR PREDICTIONS

The optimized NEC model of the biconical antenna gave impedance predictions which agreed closely with measurement and therefore it was judged to be suitable for use with antenna factor prediction.

A. Plane-Wave Antenna Factor

In order to calculate the antenna factors of the biconical antenna, the excitation applied to the NEC model was changed from a voltage source at its drive point to a plane-wave illumination having an electric-field strength equal to 1 $V \cdot m^{-1}$. The drive point of the antenna was loaded with an impedance of 50 Ω to represent a matched transmission line. The NEC simulations returned the current in this load impedance, then Ohm's law gave the voltage and (1) was used to calculate the antenna factors.

Fig. 9 shows that the antenna factor of the biconical antenna increases sharply below 70 MHz, indicating that the antenna becomes progressively less sensitive. This is due to mismatch losses, which occur because the antenna becomes capacitive.

The antenna is most sensitive at 71 MHz, where its antenna factor is 6.8 dB. Above this frequency, its antenna factor rises gently but remains less than 20 dB up to 300 MHz.

The cone resonance of the biconical antenna is shown by its predicted antenna factor at 286.3 MHz. It can be seen that

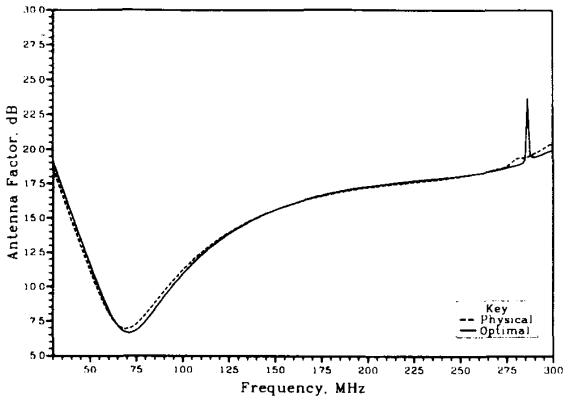


Fig. 9. Predicted antenna factors from the physical and optimized NEC models of the biconical antenna. Note that the cone resonance is not resolved by the physical model data.

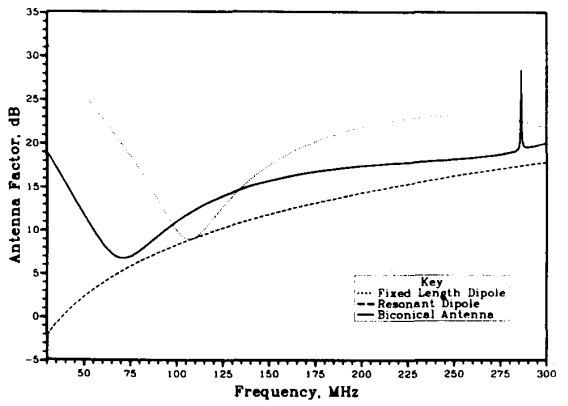


Fig. 10. Predicted antenna factor of the 1.3-m biconical antenna: a 1.3-m fixed-length dipole antenna and a resonant dipole antenna.

this desensitizes the antenna and the antenna factor becomes elevated by 9 dB. The disturbance is most pronounced over the frequency range of 285–288 MHz and does not affect the antenna significantly at other frequencies.

Fig. 9 shows that the antenna factor predictions from the physical NEC model are slightly different from those from the optimized model, although the maximum extent of the difference is only 0.7 dB. This means that it is desirable to optimize the NEC model, but using a physical model would not cause a great loss of accuracy. It also indicates that the predictions are not highly sensitive to model construction. However, the impedance predictions from the optimized model are more accurate and the difference between antenna factors from the optimized model and the physical model is comparable to measurement uncertainties.

B. Comparison of Dipole and Biconical Antennas

Fig. 10 shows predicted antenna factors for the biconical antenna and also for a fixed-length. 6.35 mm diameter dipole antenna of equal length, i.e., 1.3 m, and a resonant dipole antenna. Both of these sets of dipole antenna factors were obtained from NEC simulations of a 21 segment model antenna.

It can be seen that, above resonance, the antenna factor of the biconical antenna is between 2 and 5 dB greater than that of a resonant dipole. This means that it is less sensitive than a resonant dipole and is a consequence of the poorer match between the drive-point impedance of the biconical antenna and a 50 Ω load.

The antenna factor of the fixed-length dipole has a similar frequency dependence to that of the biconical antenna in that its antenna factor passes through a minimum close to its resonant frequency and rises either side of that. The minimum antenna factor of the dipole occurs at 107 MHz, whereas that of the biconical antenna occurs the lower frequency of 71 MHz.

Below their resonant frequencies, the fixed-length dipole and biconical antennas are capacitive and their antenna factors rise at 40 dB per decade. The fact that the resonant frequency of the biconical antenna is 36 MHz lower than than of the dipole antenna causes it to be 12 dB more sensitive at frequencies below its resonance.

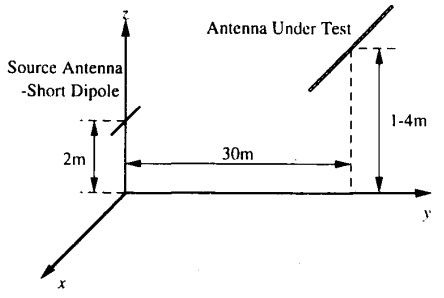


Fig. 11. NEC model used to provide a uniform illumination of the biconical antenna.

VI. Ground Plane Effects

The biconical antenna is used to measure field strength above a ground plane rather than in free space and it is important to know the extent that the antenna interacts with the ground plane in order to assess the effect that this has upon its calibration. It has been suggested that the biconical antenna interacts less than resonant dipoles with a ground plane because of its smaller size in relation to a wavelength at lower frequencies [1]; however, one recent paper seems to show the contrary [6]. We believe this is due to the use of a different formulation for the mismatch between the antenna and its feeding transmission line.

A. Uniform Illumination

A plane-wave illumination of a horizontally polarized antenna is not possible above a ground plane and therefore a NEC model was developed that allowed an appropriately uniform illumination of an antenna under test above a ground plane. Fig. 11 shows how this was achieved using a short-dipole antenna at a horizontal distance of 30 m from the antenna under test and at a height of 2 m. A voltage source was applied to the middle segment of the short dipole and thus provided illumination of the antenna under test.

The short dipole was given a length of $\lambda/20$ and a radius of $\lambda/2000$ and eleven segments were used to model it at each excitation frequency. In the absence of the antenna under test, the field uniformity was predicted over the space that would be occupied by it. This was found to be within 0.12 dB over the length of a resonant dipole at 30 MHz, but less than 0.01 dB above 100 MHz. Field uniformity was within 0.01 dB over the length of the biconical antenna and within 0.3 dB across the widest part of its cones.

B. Variation of Antenna Factors

The NEC model was first used with no antenna under test present and the field at the point where the center of the antenna under test was to be placed was predicted. 6.35-mm-diameter resonant dipoles and the biconical antenna were then placed at this position and the voltage delivered by them to a 50 Ω load was obtained. Equation (1) was then used to calculate the antenna factors at heights of 1, 2, 3, and 4 m. These values were then normalized to the previously calculated plane-wave antenna factors to yield the variations and these are shown in Fig. 12 for the resonant dipole and Fig. 13 for the biconical antenna.

1) The Resonant Dipole: Fig. 12 shows that the antenna factor of a resonant dipole above a ground plane varies about its plane-wave value in an oscillatory fashion with a decreasing amplitude as frequency increases. The amplitude of the oscillations is greater at lower heights and is considerably greater at 1 m height than at 2 m height. The antenna factor of a resonant dipole used at 1 m height falls by more than 2 dB below 45 MHz and is reduced by 4.5 dB at 30 MHz. It would be very difficult to calibrate the antenna under these conditions because of the strong dependence of antenna factor upon height.

The antenna factor of a resonant dipole used at 2 m height and below 100 MHz will vary between − 1.2 and + 1.7 dB from its plane-wave value. Above 100 MHz, these limits fall to ± 1 dB and above 200 MHz they are less than ± 0.6 dB. These limits represent the accuracy to which field strength measurement may be performed if plane-wave antenna factors are used uncorrected.

2) The Biconical Antenna: Fig. 13 shows that the antenna factor of the biconical antenna oscillates about its plane-wave value but, unlike the antenna factor of the resonant dipole, the amplitude and periodicity of these oscillations do not change steadily as functions of frequency. This is because the impedance of a resonant dipole is resistive and equal to 73 Ω, whereas the impedance of the biconical antenna varies.

As the antenna becomes capacitive below 60 MHz, antenna factor variations becomes reduced so that they have fallen to ± 1 dB at 30 MHz. The antenna becomes progressively shorter than a wavelength in this frequency range and therefore this result shows how an electrically short antenna interacts less with its surrounding environment than an antenna of appreciable length.

The minimum plane-wave antenna factor of the biconical antenna occurs around 73 MHz and when the antenna is above a ground plane, the depth of this minimum does not change significantly but its frequency shifts slightly. This gives rise

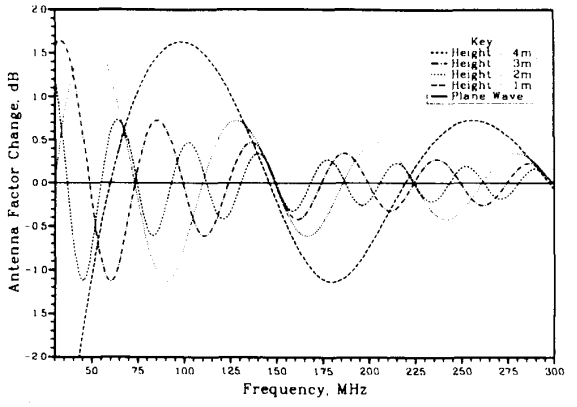


Fig. 12. Variations of the antenna factor of the resonant dipole antenna with height.

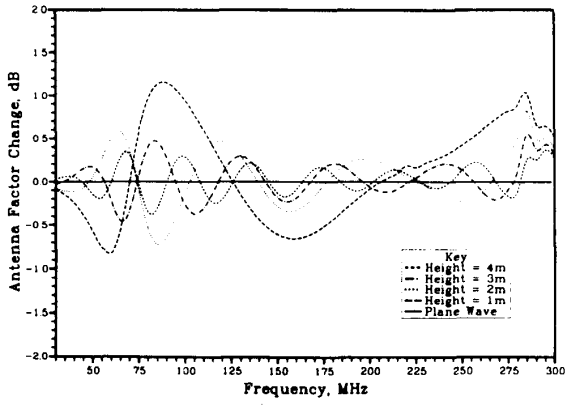


Fig. 13. Variations of the antenna factor of the biconical antenna with height.

to very small antenna factor variations at 73 MHz, but much larger ones at frequencies either side of the minimum, where the antenna factor rises steeply. The frequencies at which the antenna factor of the biconical antenna varies most are close to 60 and 80 MHz, where it varies between ±0.7 dB for an antenna at heights of 2 m and above, and between between −0.8 and +1.2 dB for an antenna at 1 m height.

In the region above 100 MHz, where the antenna has appreciable electrical length, its antenna factor varies between ±0.5 dB from its plane-wave value at heights of 2 m and above; at 1 m height these variations increase to ±0.8 dB. Close to the cone resonance at 287 MHz, the variations are slightly greater, but the antenna has already been explained to be of little use at those frequencies.

The variations of the antenna factor of the biconical antenna are generally less than those of the resonant dipole, but it is considerably more influenced by a ground plane when at 1 m height than at heights of 2 m and above. This indicates that care should be taken if it is calibrated by procedures involving measurements made at 1 m height.

C. Variation of Effective Length

The method given in Section VII was used to obtain the parameters of the Thèvenin voltage source equivalent to the

illuminated radiators of the antenna under test, i.e., the EMF induced in the antenna and its drive-point impedance. Effective length is defined as the ratio of the EMF induced in an antenna to the illuminating field and it is one of two parameters which give rise to variations in antenna factor. Equation (2) implies that for an antenna under plane wave illumination

$$AF_0 = \frac{1}{l_{e0}}\left|1 + \frac{Z_{T0}}{Z_L}\right|. \tag{4}$$

For an antenna at a general height above a ground plane

$$AF = \frac{1}{l_e}\left|1 + \frac{Z_T}{Z_L}\right| \tag{5}$$

and therefore the change in antenna factor from its plane-wave value is

$$\Delta AF = \frac{AF}{AF_0} = \frac{l_{e0}}{l_e}\left|\frac{Z_T + Z_L}{Z_{T0} + Z_L}\right| \tag{6}$$

i.e., changes in antenna factor arise from the product of changes in effective length and changes in mismatch with the load. The former of these is often ignored when calculating variations in antenna factor.

1) The Resonant Dipole: Fig. 14 shows the predicted variations in effective length of a resonant dipole antenna and it can be seen that they are much less than the variations in antenna factor shown by Fig. 12. Variations in the antenna factor of a resonant dipole therefore arise chiefly from variations in the mismatch between it and its load.

The variations in effective length are greatest at lower heights but, even at 1 m height, they are only between +0.23 and −0.16 dB. This is the maximum error that can occur if variations in the antenna factor of a resonant dipole are calculated purely from variations in drive-point impedance. This also confirms that it is valid to assume that the effective length of a dipole antenna at resonance is equal to λ/π even when the antenna is above a ground plane.

2) The Biconical Antenna: Fig. 15 shows that effective length changes of the biconical antenna are small at frequencies up to 100 MHz, and comparison with Fig. 13 indicates that antenna factor changes are primarily due to changes in mismatch. The maximum variation of effective length that occurs over this frequency range is ±0.2 dB and occurs close to the resonant frequency of 73 MHz.

At frequencies above 100 MHz, changes in the effective length of the biconical antenna become progressively larger and must be taken into account in any calculations of antenna factor variations. The variations in effective length actually become greater than the variations in antenna factor, showing that there is a tendency for variations in antenna factor due to changes in effective length to be offset by variations due to changes in mismatch.

VII. Inclusion of a Balun

The previous sections have shown how the antenna factor of the radiators of a biconical antenna may be calculated; however, this approach has not yet considered the balun of the antenna. Section IV has shown how jigs can be used to mount N-type connectors in place of the radiators of an antenna and

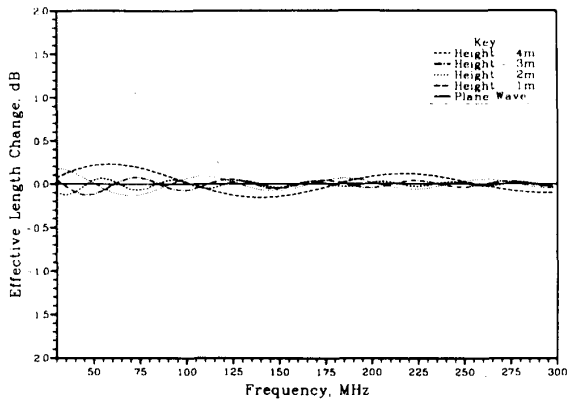


Fig. 14. Variation of the effective height of the resonant dipole antenna with height.

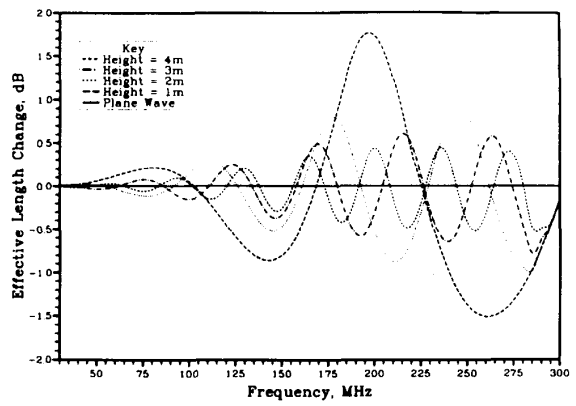


Fig. 15. Variations of the effective length of the biconical antenna with height.

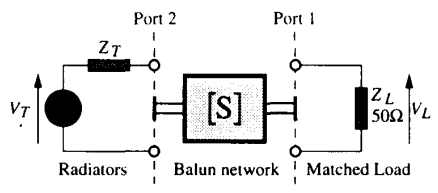


Fig. 16. Equivalent circuit of any antenna and its balun.

thereby permit calibrations to establish a reference plane at its drive point. Once this has been done the complete antenna may be represented as shown in Fig. 16.

A variation on the impedance measurement approach was used to measure the scattering parameters of the balun of the Schwarzbeck biconical antenna and NEC was used to derive the voltage source parameters of its illuminated radiators.

A. Measurement of Balun Scattering Parameters

The two jigs shown in Fig. 5 were mounted upon the antenna support and the network analyzer was calibrated to measure the reflection coefficient at the usual coaxial connector of the antenna. Pairs of open, short, and matched loads were connected to the N-type connectors on the jigs and the

corresponding reflection coefficients at the antenna connector were then Γ_{oc}, Γ_{sc}, and Γ_{mt}. The general expression for the reflection coefficient of a reciprocal two-port network terminated by a load on its second port is

$$\Gamma_{in} = S_{11} + \frac{S_{12}^2}{1 - S_{22}\Gamma_L} \tag{7}$$

where S_{11}, S_{12}, and S_{22} are the scattering parameters of the two-port network and Γ_L is the reflection coefficient of the load. For the three above-mentioned load conditions, it is possible to show that

$$S_{11} = \Gamma_{mt} \tag{8}$$

$$S_{12} = \sqrt{\frac{2(\Gamma_{mt} - \Gamma_{oc})(\Gamma_{mt} - \Gamma_{sc})}{(\Gamma_{sc} - \Gamma_{oc})}} \tag{9}$$

$$S_{22} = \frac{2\Gamma_{mt} - \Gamma_{oc} - \Gamma_{sc}}{\Gamma_{sc} - \Gamma_{oc}}. \tag{10}$$

A phase correction corresponding to a free-space distance of 89 mm was applied to S_{12} and S_{22} in order to move the reference plane of port 2 to the drive point of the antenna. It should be noted that because port 2 was formed from two 50 Ω ports in series, its characteristic impedance was 100 Ω.

B. Equivalent Voltage Source

The NEC model of the biconical antenna was illuminated by a 1 V · m^{-1} plane wave and two simulations were run; one to return the current delivered by the antenna to a 50 Ω load and the other to return the current delivered to a 100 Ω load. From these two currents it was possible to calculate the EMF developed in the antenna,, V_T, and the drive-point impedance, Z_T.

$$V_T = \frac{I_{L1}I_{L2}(Z_{L2} - Z_{L1})}{I_{L1} - I_{L2}} \tag{11}$$

$$Z_T = \frac{I_{L2}Z_{L2} - I_{L1}Z_{L1}}{I_{L1} - I_{L2}} \tag{12}$$

where I_{L1} is the current delivered to Z_{L1} and I_{L2} is the current delivered to Z_{L2}.

C. Calculation of Antenna Factor

The input impedance of port 2 of the balun is given by

$$Z_{in2} = Z_{02}\frac{1 + S_{22}}{1 - S_{22}}. \tag{13}$$

The input voltage at port 2 is therefore

$$V_{in2} = V_T\frac{Z_{in2}}{Z_T + Z_{in2}}. \tag{14}$$

The reverse voltage gain of a two-port network (i.e., port 2 to port 1) with its output port matched is given by

$$A_V = \sqrt{\frac{Z_{01}}{Z_{02}}}\frac{S_{12}}{1 + S_{12}}. \tag{15}$$

The received voltage with the antenna is given by the voltage gain of the balun multiplied by the input voltage to the balun,

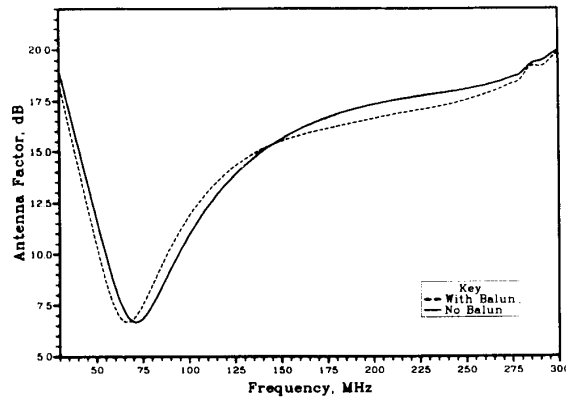


Fig. 17. Predicted plane-wave antenna factors for the biconical antenna with and without its balun.

and therefore combining (13)–(15) gives

$$V_L = V_T\frac{S_{12}\sqrt{Z_{01}Z_{02}}}{Z_T(1 - S_{22}) + Z_{02}(1 + S_{22})}. \tag{16}$$

Substituting this into (1) gives the expression for the antenna factor of the biconical antenna incorporating its balun.

$$AF = 20\log_{10}\left|\frac{E}{V_T}\cdot\frac{Z_T(1 - S_{22}) + Z_{02}(1 + S_{22})}{S_{12}\sqrt{Z_{01}Z_{02}}}\right|. \tag{17}$$

The characteristic impedances Z_{01} and Z_{02} were equal to 50 Ω and 100 Ω, respectively, for the biconical antenna.

D. Predicted Antenna Factors

Predicted antenna factors were obtained from (2) for the biconical antenna with no balun and from (17) with its balun taken account of. Fig. 17 shows these data.

Fig. 17 shows that the balun not only provides a balanced-to-unbalanced transformation but it also modifies the antenna factor appreciably. This is because it acts as an impedance transformer and thus changes the effective load impedance presented to the radiators.

Baluns are usually expected to introduce a small loss at all frequencies when they are used with resonant dipoles but Fig. 17 shows that they have a more complicated effect on the biconical antenna, and can either increase or reduce its antenna factor according to the precise frequency of excitation. Manufacturers' balun loss data should therefore not be used to correct antenna factors for biconical antennas.

The balun increases the sensitivity of the antenna by 1.2 dB below 69 MHz; reduces sensitivity by up to 1.1 dB between 69 and 145 MHz, and improves sensitivity by up to 0.7 dB between 145 and 300 MHz.

VIII. Conclusions

NEC can be used to accurately model the radiating structures of a wire biconical antenna, although care has to be taken when modeling the excitation region. If the region between the cones is modeled by a simple cone-linking wire, five segments should be used for this with excitation or loading applied to the middle one. It is also advantageous to use a

tapered segmentation scheme to model the cone wires so that all segments in the excitation region have similar lengths.

Comparison of measured and predicted drive-point impedances can be used to gauge the accuracy of an NEC model to be used for antenna factor prediction because impedance is easier to measure than the antenna factor and also more sensitive to bad model design. Models of the biconical antenna in this paper have given accurate impedance predictions except at low frequencies where they have been approximately 16% too capacitive. This is believed to be because they do not model the bulky metal supports at the base of either cone and a model would have to incorporate these to give better predictions.

Antenna factors vary from their plane-wave values when an antenna is above a ground plane. These variations may be calculated purely from variations in the impedance of the resonant dipole, but effective length changes must also be considered with the biconical antenna. Variations for the biconical antenna are less than for the resonant dipole; and for antennas at 2-m height, they are between -1.2 and $+1.7$ dB for the dipole and between ± 0.7 dB for the biconical antenna.

A combination of NEC modeling and measurement of the two-port network parameters of a balun can be used to predict the antenna factors of a complete real biconical antenna. In order to be able to measure the network parameters of the balun and validate the NEC simulation through impedance measurement it is necessary to be able to define a reference plane at the drive point of the radiators of the antenna. This is only possible if the balun of the antenna has a metal casing through which coaxial penetrations feed each radiator. It will be very difficult to use simulation to predict the antenna factors of biconical antennas with their feed regions enclosed in plastic cases.

REFERENCES

[1] A. A. Smith *et al.*, "Calculation of site attenuation from antenna factors," *IEEE Trans. Electromagn. Compat.*, vol. EMC-24, no. 3, pp. 301–316, Aug. 1982.

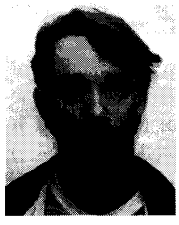
[2] A. A. Smith *et al.*, "Standard site method for antenna calibration," *IEEE Trans. Electromagn. Compat.*, vol. EMC-24, no. 3, pp. 316–322, Aug. 1982.

[3] A. Sugiura, "Correction factors for normalized site attenuation," *IEEE Trans. Electromagn. Compat.*, vol. 34, no. 4, pp. 461–470. Nov. 1992.

[4] G. J. Burke and A. J. Poggio, "Numerical electromagnetics code (NEC)—Method of moments," Rep. UICD-18834, Livermore Nat. Lab., Livermore, CA.

[5] S. M. Mann and A. C. Marvin, "A computer study of the calibration of the skeletal biconical antenna and the resonant dipole antenna," presented at an IEE colloquium on "Radiated Emission test Facilities," London, UK, June 2, 1992.

[6] B. A. Austin and A. P. C. Fourie, "Characteristics of the wire biconical antenna used for EMC measurements," *IEEE Trans. Electromagn. Compat.*, vol. 33, no. 3, pp. 179–187, Aug. 1991.



S. M. Mann was born in Coventry, England, on November 15, 1965. He received the B.Sc. degree in electronics in 1988, and the D.Phil. degree in electromagnetic compatibility in 1993, both from the University of York, York, England.

Between 1988 and 1993 he worked in the EMC research group of the Department of Electronics at the University of York, first as a Research Assistant and as a Research Fellow from 1992. He now works as a Higher Scientific Officer in the RF Field Group of the Non-Ionising Radiation Department at the National Radiological Protection Bourd, Chilton, Didcot, OXON, OX11 0RQ, United Kingdom. His interests are concerned with computer modeling of antennas for performance evaluation, calibration, and hazard assessment purposes; measurement and computational dosimetry techniques for the assessment of biological interactions with fields; and the characterization of EMC test environments.

Andrew C. Marvin (M'85) is a Senior Lecturer and Head of the Electromagnetic Compatibility Research Group at the University of York, York, England. His wide-ranging interests include calibration aspects of EMC measurements in screened rooms, computer applications to EMC design, EMC education, and antenna design